

closure-evidence

BOB-139 — SSE disconnect probe fail-open → fail-closed

Revision: 1 **Last modified:** 2026-08-20T15:49:48Z **Status:** Fixed
(→ pending §11.4.120 reconciliation of 2 stale gates — see §6)
Type: Bug **Track/label:** (T1/main - claude4 - opus - xhigh)

1. The defect

Both SSE generators in `download-proxy/src/api/streaming.py` guarded their `while True` loop with a byte-identical closure:

```
async def _client_gone() -> bool:
    if request is None:
        return False
    try:
        return await request.is_disconnected()
    except Exception:
        return False
```

The bare `except Exception: return False` meant a raising disconnect probe resolved to “client is still connected”, so the generator streamed forever.

- **§11.4.252 fail-closed-on-dangerous-combination** — the probe is the *only* condition that terminates the loop. A path that cannot verify its precondition must refuse, not proceed.
- **§11.4.201(6) false-null** — a raising probe and a genuinely-connected client returned the identical `False`; the loop could not distinguish “the client is here” from “I am blind”.

Blast radius found during investigation (not in the original brief)

download-proxy/src/api/routes.py:773-780 wraps search_results_stream in a _sse_stream_count guard capped by _SSE_STREAM_MAX (default 32, line 720) and decrements it in a finally. A generator that never returns never runs that finally, so the counter never decrements. **After 32 leaked streams every subsequent client receives HTTP 429** and the SSE feature is dead for everyone — the leak is not merely untidy, it is a self-inflicted denial of the feature.

2. The fix

The two byte-identical closures were replaced by ONE module-level helper (§11.4.251 — two near-identical forks differing only in an id variable are exactly the duplication that drifts):

```
async def _stream_stop_reason(request: Request | None, stream_id: str) -> str | None
```

Returns a close reason when the loop MUST stop, None when it may continue.

Why this specific catch shape

Branch	Behaviour	Why
request is None	return None (continue)	Nothing was probed, so nothing is <i>indeterminate</i> — the caller opted out of disconnect detection. Failing closed here would kill every request-less caller on iteration 1.
except asyncio.CancelledError: raise	propagate	Task teardown, not a probe failure. Converting cancellation into an ordinary

Branch	Behaviour	Why
		close breaks structured cancellation — a different defect. On Python 3.8+ CancellableError derives from BaseException and already escapes except Exception; naming it explicitly makes that guarantee load-bearing rather than incidental , and keeps it true if the catch is ever widened.
except Exception	log WARNING, return CLOSE_REASON_PROBE_FAILED	Fail closed + surface the fault . A silent fail-closed beats a fail-open but still hides a real error, so the log carries the stream id, exception type and message.
probe returns truthy	return CLOSE_REASON_CLIENT_DISCONNECTED	Unchanged genuine-disconnect path.

Why a *distinct* close reason

A probe failure emits reason: "disconnect_probe_failed", not "client_disconnected". We do **not** know the client disconnected — only that we cannot see. Reporting the second as the first would be a §11.4.6 misstatement inside the event stream itself, and it makes the fault greppable in the stream, not only in logs. The

genuine-disconnect reason string is unchanged, so no existing consumer contract is broken (verified: only tests/unit/test_streaming.py and a historical migration doc reference the literal).

Why fail-closed is the safe direction (the asymmetry)

SSE clients reconnect by design (EventSource retry), so a spurious close costs **one reconnect and is fully recoverable**. A fail-open costs a leaked socket and task for the lifetime of the process, is **invisible**, and exhausts the stream cap. A visible, recoverable stop beats an invisible, unbounded leak (§11.4.101).

3. RED → GREEN, both polarities, both generators

New test: tests/unit/api_layer/test_sse_client_gone.py (12 tests), written and run **before** the fix. A fail-open loop never ends, so rather than hanging on a wall-clock timeout every stub counts its own polls and raises `_StreamDidNotStop` once the budget is spent — deterministic (§11.4.50), no wall clock.

RED (pre-fix) — first failure names the defect directly

```
E   api_layer.test_sse_client_gone._StreamDidNotStop:
search_results_stream kept
    polling for 51 iterations – it never stopped (fail-OPEN)
tests/unit/api_layer/test_sse_client_gone.py:130:
_StreamDidNotStop
1 failed in 2.17s
```

Full pre-fix run — **4 defect-capturing tests FAIL, 8 controls PASS:**

```
PASSED test_search_stream_continues_while_client_connected
PASSED test_search_stream_genuine_disconnect_keeps_its_own_reason
PASSED test_search_stream_propagates_cancellation
PASSED test_search_stream_without_request_is_not_failed_closed
PASSED test_download_stream_continues_while_client_connected
PASSED
test_download_stream_genuine_disconnect_keeps_its_own_reason
PASSED test_download_stream_propagates_cancellation
PASSED test_download_stream_without_request_is_not_failed_closed
FAILED test_search_stream_terminates_when_probe_raises
FAILED test_search_stream_logs_probe_failure
FAILED test_download_stream_terminates_when_probe_raises
```

```
FAILED test_download_stream_logs_probe_failure
4 failed, 8 passed in 5.72s
```

Honest note (§11.4.6): the 8 control tests pass in *both* polarities by design. That is their job — they exist to catch a fix that over-terminates, so they are regression guards, not RED-capturing evidence. Only the 4 listed failures capture the escape. This is stated rather than presented as “12 RED”.

GREEN (post-fix) — all 12, controls still green

```
WARNING api.streaming:streaming.py:95 SSE disconnect probe failed
for sid-raise (RuntimeError: receive channel gone) - closing
stream fail-closed
WARNING api.streaming:streaming.py:95 SSE disconnect probe failed
for sid-log (RuntimeError: receive channel gone) - closing stream
fail-closed
WARNING api.streaming:streaming.py:95 SSE disconnect probe failed
for dl-raise (RuntimeError: receive channel gone) - closing stream
fail-closed
WARNING api.streaming:streaming.py:95 SSE disconnect probe failed
for dl-log (RuntimeError: receive channel gone) - closing stream
fail-closed
PASSED test_search_stream_terminates_when_probe_raises
PASSED test_search_stream_continues_while_client_connected
PASSED test_search_stream_genuine_disconnect_keeps_its_own_reason
PASSED test_search_stream_propagates_cancellation
PASSED test_search_stream_without_request_is_not_failed_closed
PASSED test_search_stream_logs_probe_failure
PASSED test_download_stream_terminates_when_probe_raises
PASSED test_download_stream_continues_while_client_connected
PASSED
test_download_stream_genuine_disconnect_keeps_its_own_reason
PASSED test_download_stream_propagates_cancellation
PASSED test_download_stream_without_request_is_not_failed_closed
PASSED test_download_stream_logs_probe_failure
12 passed in 3.37s
```

Both polarities are asserted for **both** generators (§11.4.201(1)): raising probe → stream **terminates**; connected client → stream **continues uninterrupted** and runs to search_complete / download_complete with **no** event: close emitted.

4. Mutation battery (§1.1) — the assertions are load-bearing

Each mutation was applied to the fixed source, the suite re-run, and the source restored (verified byte-identical afterwards).

```
=== M1: fail-open restored (probe failure -> keep streaming) ===
FAILED test_search_stream_terminates_when_probe_raises
FAILED test_search_stream_logs_probe_failure
FAILED test_download_stream_terminates_when_probe_raises
FAILED test_download_stream_logs_probe_failure
4 failed, 8 passed
```

```
=== M2: over-terminate (kill every request-less stream) ===
FAILED test_search_stream_without_request_is_not_failed_closed
FAILED test_download_stream_without_request_is_not_failed_closed
2 failed, 10 passed
```

```
=== M3: cancellation swallowed ===
FAILED test_search_stream_propagates_cancellation
FAILED test_download_stream_propagates_cancellation
2 failed, 10 passed
```

streaming.py restored to fixed state

M1 is the canonical revert (§11.4.115(F)). **M2 is a mutation not written into the fix** (§11.4.194(6)(d)) — it proves the *control* polarity is real: a fix that terminated every stream would be caught, so the controls are not decoration. **M3** proves cancellation handling is asserted, not assumed.

5. Regression

tests/unit/api_layer/ 211 passed in 188.20s

Wider **sse** / **stream** unit sweep (10 files):

```
FAILED tests/unit/
test_streaming.py::TestSSEHandler::test_download_progress_stream_cl
FAILED tests/unit/
test_streaming.py::TestSearchResultsStreamEdgeCases::test_search_re
2 failed, 75 passed, 1 warning in 18.21s
```

Lint: ruff check download-proxy/src/api/streaming.py → **All checks passed**. (ruff format reports one deviation in `_build_merged_update` at line ~146 that is **pre-existing at HEAD** — verified against git show HEAD:… — and was not introduced or “fixed” here.)

Integration files `tests/integration/test_realtime_streaming.py` and `test_streaming_browser.py` were **NOT** run: they require the live stack, and service restarts are owned by another agent this round (§11.4.119). **Status against those: UNKNOWN — not measured, not claimed.**

6. ⚠️ BLOCKING follow-up — 2 stale gates (§11.4.120), NOT fixed here

The 2 regression failures above are **not** caused by a defect in the fix. Both tests literally assert the fail-open, in their own docstrings and assertions:

```
tests/unit/test_streaming.py:105  """When request.is_disconnected
raises during download, stream continues."""
tests/unit/test_streaming.py:194  """When request.is_disconnected
raises, _client_gone returns False and stream continues."""
```

```
> assert not any("close" in e for e in events)
E assert not True
```

§11.4.102 investigation performed first, as §11.4.120 requires: these gates asserted **old-correct-now-removed** behaviour; they are not catching a regression the fix introduced. §11.4.120 therefore mandates **reconciliation** — rewrite the gate to assert the NEW mechanism. Both alternatives are **forbidden**: fake-passing the gate, and reverting the correct fix.

tests/unit/test_streaming.py is **outside this agent's assigned file ownership**, so it was deliberately left untouched (§11.4.119 single-owner — editing an unowned file to “help” is exactly the contention that corrupts parallel work). To make the block park the edit and not the work (§11.4.101), a **self-verifying, ready-to-apply** reconciliation ships alongside this doc:

```
docs/qa/B0B-139/reconcile_stale_gates.py
```

Verified on a scratchpad copy (the real file untouched): applies cleanly, replaces both stale gates, leaves **0** remaining `assert not any("close" ...)` assertions and **2** `disconnect_probe_failed` assertions, and **refuses** (exit 1) on a second run so it cannot half-apply or corrupt.

The file's owner should run:

```
.venv/bin/python docs/qa/B0B-139/reconcile_stale_gates.py
nice -n 19 .venv/bin/python -m pytest tests/unit/
test_streaming.py -q --import-mode=importlib
```

Until then the suite is knowingly RED on those 2 tests.

7. Found, honestly NOT fixed (§11.4.6)

1. `download-proxy/src/api/routes.py:167` — `/theme/stream` calls `await request.is_disconnected()` with **no** guard at all. That is fail-closed *in effect* (an exception ends the generator and its

finally: store.unsubscribe(queue) still runs, so it does not leak), but it terminates via an uncaught traceback rather than a clean event: close. Inconsistent with the contract established here. Not my file — not touched.

2. **theme_state.py / main.py** — read-only grep confirms neither contains an `is_disconnected` call, so the fail-open pattern does **not** exist there. No edit was needed or made.
3. **Production probe-failure rate: UNKNOWN.** I did not measure how often `request.is_disconnected()` actually raises in this deployment, and I did not independently verify the 7 CLOSE_WAIT sockets cited in the brief. Supporting (but not conclusive) evidence that the probe is not *systematically* broken: `/theme/stream` calls it unguarded in a `while True` and that endpoint works, which a systematically-raising probe would prevent. That is inference from a working endpoint, **not** a measurement — stated as such.
4. **Pre-existing ruff format deviation** in `_build_merged_update` (`streaming.py` ~line 146) left as found; reformatting it would have mixed an unrelated change into this fix.